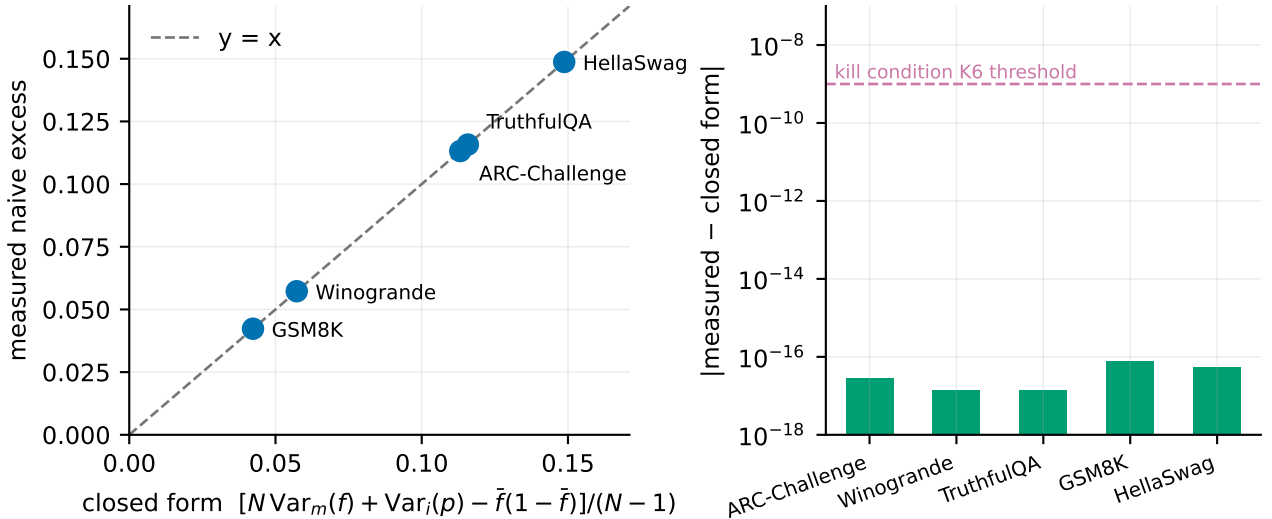
a Lemma 2 predicts the reported excess exactly
y = x
HellaSwag
TruthfulQA
ARC-Challenge
Winogrande
GSM8K
measured naive excess
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.00
0.05
0.10
0.15
closed form $[N\,\mathrm{Var}_m(f) + \mathrm{Var}_i(p) - \bar{f}(1-\bar{f})]/(N-1)$
b Residuals at machine precision
kill condition K6 threshold
$|\text{measured} - \text{closed form}|$
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
10^{-18}
ARC-Challenge
Winogrande
TruthfulQA
GSM8K
HellaSwag